MurderAssaultCase-R-Script.R

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```
USArrests <- read.csv("D:/IIM Sirmaur Data Analytics Course/MurderAssaultCase.csv",
                      stringsAsFactors=TRUE,row.names = 1) #

#rm(list = ls())

summary(USArrests)
```

```
##      Murder          Assault         UrbanPop          Rape
##  Min.   : 0.800   Min.   : 45.0   Min.   :32.00   Min.   : 7.30
##  1st Qu.: 4.075   1st Qu.:109.0   1st Qu.:54.50   1st Qu.:15.07
##  Median : 7.250   Median :159.0   Median :66.00   Median :20.10
##  Mean   : 7.788   Mean   :170.8   Mean   :65.54   Mean   :21.23
##  3rd Qu.:11.250   3rd Qu.:249.0   3rd Qu.:77.75   3rd Qu.:26.18
##  Max.   :17.400   Max.   :337.0   Max.   :91.00   Max.   :46.00
```

```
str(USArrests)
```

```
## 'data.frame':    50 obs. of  4 variables:
##  $ Murder  : num  13.2 10 8.1 8.8 9 7.9 3.3 5.9 15.4 17.4 ...
##  $ Assault : int  236 263 294 190 276 204 110 238 335 211 ...
##  $ UrbanPop: int  58 48 80 50 91 78 77 72 80 60 ...
##  $ Rape    : num  21.2 44.5 31 19.5 40.6 38.7 11.1 15.8 31.9 25.8 ...
```

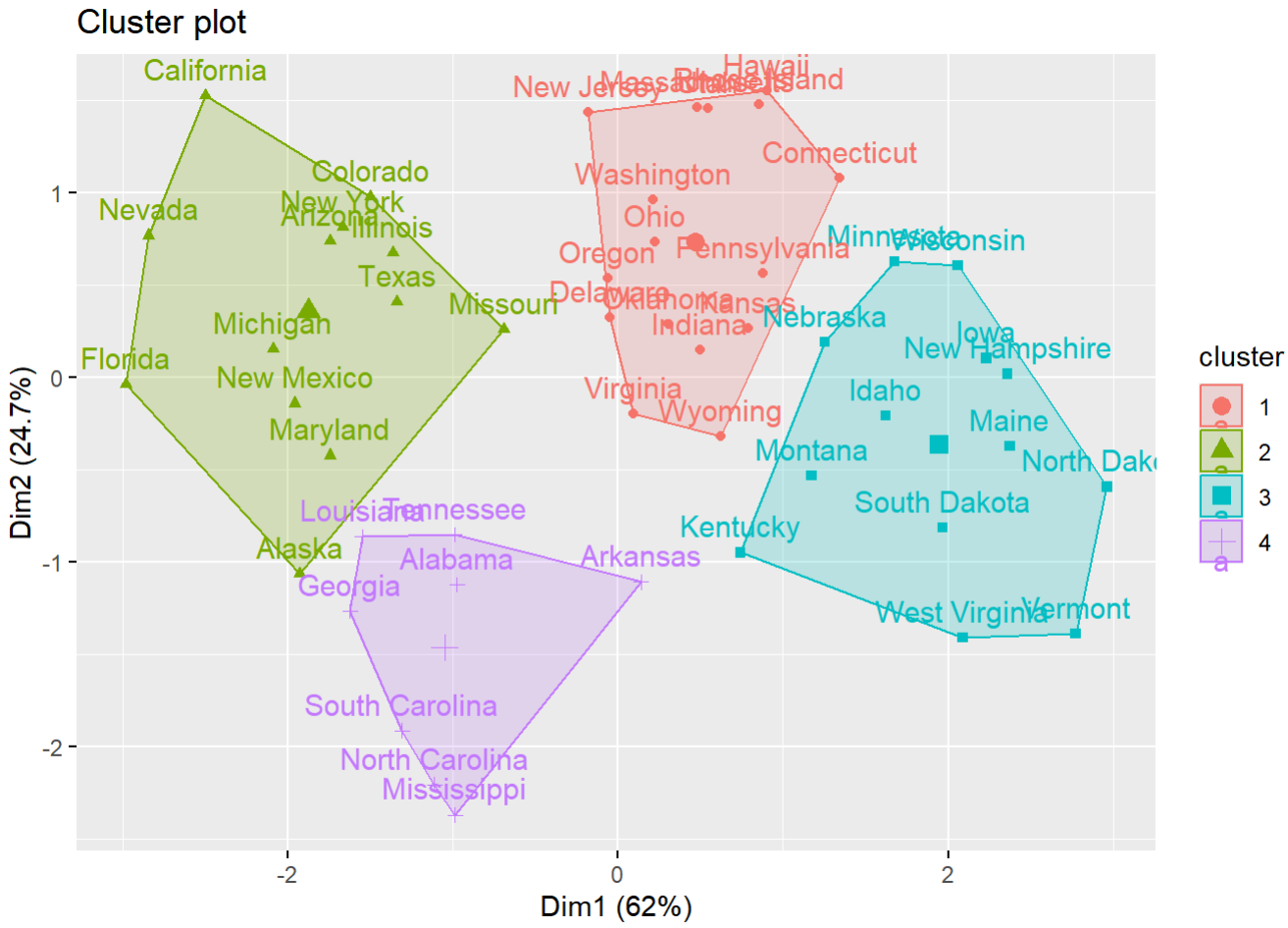
```
#install Cluster package and Factoextra package
library(cluster)
library(factoextra)
```

```
## Loading required package: ggplot2
```

```
## Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WBa
```

```
USArrests1 = scale(USArrests) #Scaling or Normalization of Data to make them similarly close
#apply Scaling before using k means
km = kmeans(USArrests1,4) # k means clustering where we selected no of clusters and pass that
in dataset

fviz_cluster(km,data = USArrests1)
```




```
#makes a plot of 2 clusters in USArrest
str(km)
```

```
## List of 9
##  $ cluster     : Named int [1:50] 4 2 2 4 2 2 1 1 2 4 ...
##   ..- attr(*, "names")= chr [1:50] "Alabama" "Alaska" "Arizona" "Arkansas" ...
##  $ centers     : num [1:4, 1:4] -0.489 0.695 -0.962 1.412 -0.383 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : chr [1:4] "1" "2" "3" "4"
##   .. ..$ : chr [1:4] "Murder" "Assault" "UrbanPop" "Rape"
##  $ totss       : num 196
##  $ withinss    : num [1:4] 16.21 19.92 11.95 8.32
##  $ tot.withinss: num 56.4
##  $ betweenss   : num 140
##  $ size        : int [1:4] 16 13 13 8
##  $ iter        : int 2
##  $ ifault      : int 0
##  - attr(*, "class")= chr "kmeans"
```

```
Accuracy = km$betweenss/km$totss

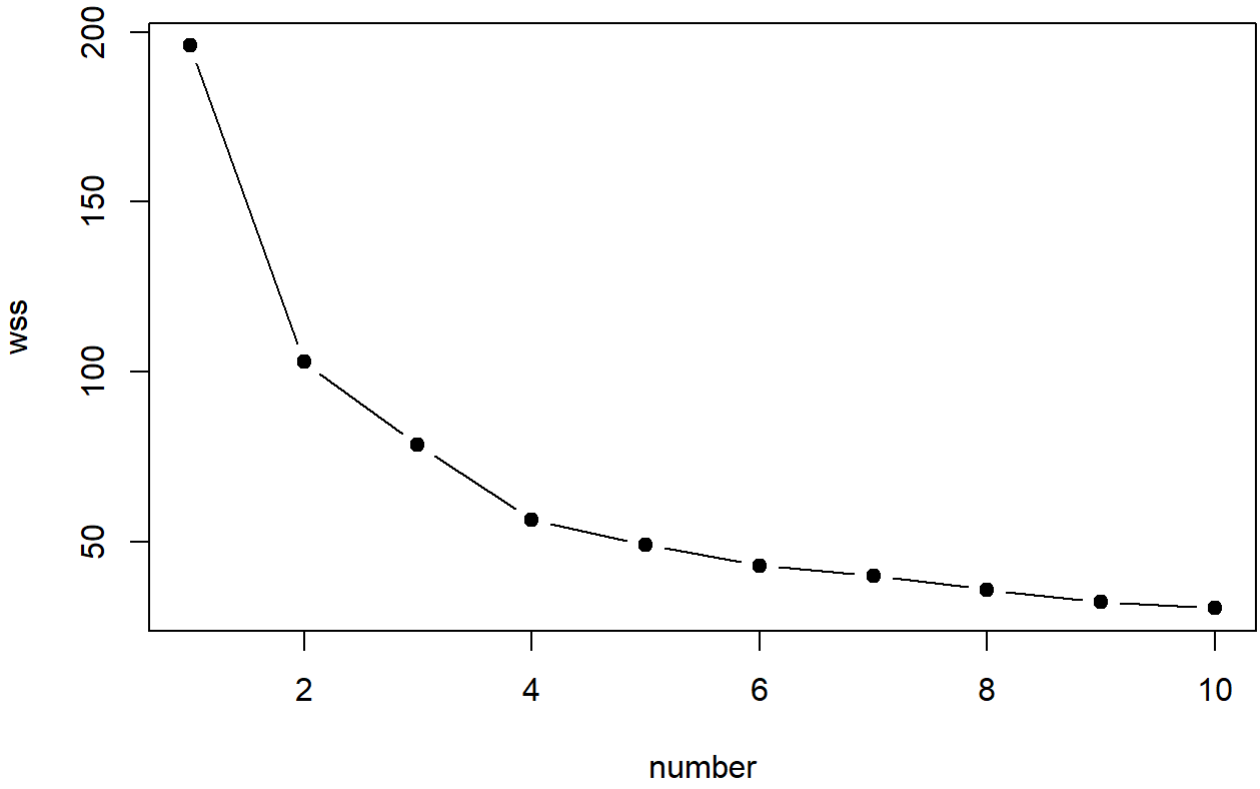
##Totss=betweens (Higher)+Withness(lower)
#Accuracy == 70%(good clustering)
Accuracy
```

```
## [1] 0.7122287
```

```r
#Elbow Method to improve accuracy
number = 1:10
wss = 1:10

for(i in 1:10)
{
  wss[i]=kmeans(USArrests1,i)$tot.withinss


}
plot(number,wss,type="b",pch=19)
```




```r
#Will create a plot where we will select no of clusters
#on point 4 an elbow is forming, that's our optimal no of clusters

#Profiling of Data
USArrests$cluster = km$cluster

cmeans = aggregate(USArrests,by=list(USArrests$cluster),mean)
cmeans
```

```
##   Group.1   Murder   Assault UrbanPop     Rape cluster
## 1       1  5.65625 138.87500 73.87500 18.78125       1
## 2       2 10.81538 257.38462 76.00000 33.19231       2
## 3       3  3.60000  78.53846 52.07692 12.17692       3
## 4       4 13.93750 243.62500 53.75000 21.41250       4
```